Chapter 8 — Physical Optics and Gravitational Waves

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. Polarization proves light is:

(a) longitudinal **(b)** stationary **(c)** transverse **(d)** mechanical

2. Polarization restricts vibrations to:

(a) random directions **(b)** direction of travel **(c)** all planes **(d)** one plane

3. Longitudinal waves:

(a) are always polarized **(b)** have no frequency **(c)** can be polarized **(d)** cannot be polarized

4. A polaroid transmits:

(a) one vibration direction **(b)** no light **(c)** only sound **(d)** all directions equally

5. Two crossed polaroids transmit:

(a) half light always **(b)** double light **(c)** maximum light **(d)** no light

6. Malus's law is:

(a) $I = I_0 \sin^2\theta$ **(b)** $I = I_0 \cos\theta$ **(c)** $I = I_0 \cos^2\theta$ **(d)** $I = I_0 \tan^2\theta$

7. The amplitude passed by analyzer is:

(a) $A_0 \cos\theta$ **(b)** $A_0 \sin\theta$ **(c)** $A_0 \tan\theta$ **(d)** $A_0 \cos^2\theta$

8. Brewster's law is:

(a) $\tan\theta_p = 1/n$ **(b)** $\cos\theta_p = n$ **(c)** $\sin\theta_p = n$ **(d)** $\tan\theta_p = n$

9. At Brewster's angle, reflected and refracted rays are:

(a) 0° apart **(b)** 45° apart **(c)** parallel **(d)** 90° apart

10. If n = √3, polarizing angle is:

(a) 60° **(b)** 90° **(c)** 30° **(d)** 45°

11. Optically active substances rotate:

(a) frequency **(b)** plane of polarization **(c)** wavelength only **(d)** speed of light in vacuum

12. Polaroids in sunglasses reduce:

(a) sound **(b)** gravity **(c)** heat only **(d)** glare

13. Sky light is partially polarized due to:

(a) diffraction only **(b)** scattering **(c)** refraction only **(d)** reflection only

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14. Gravitational waves were predicted by:

(a) Newton **(b)** Maxwell **(c)** Galileo **(d)** Einstein

15. Gravitational waves travel at:

(a) zero speed **(b)** speed of light **(c)** half light speed **(d)** speed of sound

16. Gravitational waves are:

(a) ripples in space-time **(b)** ocean waves **(c)** sound waves **(d)** electromagnetic waves

17. The first direct detection of gravitational waves was in:

(a) 2015 **(b)** 2012 **(c)** 1916 **(d)** 2017

18. The 2015 gravitational wave signal came from:

(a) the Sun **(b)** two neutron stars **(c)** two black holes **(d)** a supernova

19. The first neutron-star gravitational wave detection was in:

(a) 2015 **(b)** 2016 **(c)** 2017 **(d)** 2012

20. LIGO uses:

(a) optical lenses only **(b)** radio telescopes **(c)** laser interferometry **(d)** sonar

21. LIGO arms are each:

(a) 1 km **(b)** 10 km **(c)** 4 km **(d)** 3 km

22. Virgo arms are each:

(a) 10 km **(b)** 4 km **(c)** 1 km **(d)** 3 km

23. The 2017 Nobel Prize for gravitational waves went to:

(a) Higgs and Englert **(b)** Einstein only **(c)** Curie and Rutherford **(d)** Weiss, Barish and Thorne

24. Nergis Mavalvala is associated with:

(a) LIGO **(b)** radioactivity **(c)** X-ray diffraction **(d)** Higgs discovery

25. Continuous gravitational waves come from:

(a) binary mergers only **(b)** supernovae only **(c)** spinning neutron stars **(d)** random background

26. Compact binary inspiral waves come from:

(a) sound sources **(b)** single stationary star **(c)** planets only **(d)** orbiting compact objects

27. Burst gravitational waves come from:

(a) violent events like supernovae **(b)** stationary masses **(c)** steady spinning only **(d)** random background only

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28. Stochastic gravitational waves are:

(a) burst only **(b)** single binary only **(c)** random background from many sources **(d)** continuous only

29. Gravitational wave strain is:

(a) dimensionless **(b)** measured in joules **(c)** measured in metres **(d)** measured in newtons

30. Typical LIGO strain is about:

(a) 10^{-21} **(b)** 10^{-3} **(c)** 10^{-1} **(d)** 10^{3}

31. A light year is about:

(a) 9.5×10^{12} km **(b)** 9.5×10^{9} km **(c)** 9.5×10^{6} km **(d)** 9.5×10^{15} km

32. If gravitational wavelength is 3000 km, frequency is about:

(a) 100 Hz **(b)** 10 Hz **(c)** 1000 Hz **(d)** 1 Hz

33. If wavelength is 4000 km, frequency is:

(a) 750 Hz **(b)** 100 Hz **(c)** 75 Hz **(d)** 7.5 Hz

34. The analyzer is used to:

(a) produce sound **(b)** examine polarized light **(c)** measure temperature **(d)** measure mass

35. Maximum transmitted intensity occurs when axes are:

(a) perpendicular **(b)** at 60° **(c)** parallel **(d)** at 45°

36. If axes are at 60°, transmitted fraction is:

(a) 1 **(b)** 0.25 **(c)** 0.75 **(d)** 0.5

37. If intensity is reduced by 85%, transmitted fraction is:

(a) 0.25 **(b)** 0.15 **(c)** 0.5 **(d)** 0.85

38. Unpolarized light through one polaroid gives intensity:

(a) I_0 **(b)** $I_0/2$ **(c)** zero **(d)** $I_0/4$

39. Polarized light of 100 W/m² at 45° gives:

(a) 100 W/m² **(b)** 50 W/m² **(c)** 25 W/m² **(d)** 75 W/m²

40. Amplitude 10 units at 30° gives transmitted amplitude:

(a) 8.66 units **(b)** 5 units **(c)** 10 units **(d)** 7.07 units

41. For glass n = 1.5, polarizing angle is about:

(a) 45° **(b)** 56.3° **(c)** 33.7° **(d)** 60°

42. For water n ≈ 1.33, polarizing angle is about:

(a) 37° **(b)** 45° **(c)** 60° **(d)** 53°

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43. Stress analysis uses polaroids because stressed material becomes:

(a) conductive **(b)** magnetic **(c)** birefringent **(d)** radioactive

44. Without atmosphere, sky would appear:

(a) green **(b)** red **(c)** dark **(d)** blue

45. Gravitational waves pass through matter:

(a) reflected fully **(b)** completely absorbed **(c)** almost unchanged **(d)** slowed to sound speed

46. Tidal forces from gravitational waves cause:

(a) stretching and compressing **(b)** heating only **(c)** ionization only **(d)** magnetization only

47. Every accelerating mass produces:

(a) gravitational waves **(b)** only sound **(c)** only heat **(d)** only light

48. The primary method to detect gravitational waves is:

(a) photography **(b)** laser interferometry **(c)** sonar **(d)** radar

49. A binary black hole merger is a source of:

(a) gravitational waves **(b)** only sound **(c)** only radio waves **(d)** only X-rays

50. The phenomenon of polarization is associated with:

(a) only water **(b)** only sound **(c)** transverse waves **(d)** longitudinal waves

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Answer Key with Brief Reasons

1. Answer: (c) transverse — Only transverse waves can be polarized.

2. Answer: (d) one plane — Plane-polarized light vibrates in one plane.

3. Answer: (d) cannot be polarized — Vibrations are along travel.

4. Answer: (a) one vibration direction — It acts as a polarizer.

5. Answer: (d) no light — Axes at 90° block light.

6. Answer: (c) $I = I_0 \cos^2\theta$ — Intensity follows amplitude squared.

7. Answer: (a) $A_0 \cos \theta$ — Component along axis.

8. Answer: (d) $\tan \theta_p = n$ — Polarizing angle relation.

9. Answer: (d) 90° apart — They are perpendicular.

10. Answer: (a) 60° — $\tan \theta = \sqrt{3}$.

11. Answer: (b) plane of polarization — Quartz and sugar solution rotate plane.

12. Answer: (d) glare — They block polarized reflected light.

13. Answer: (b) scattering — Atmospheric scattering polarizes light.

14. Answer: (d) Einstein — Einstein predicted them in 1916.

15. Answer: (b) speed of light — They propagate at c.

16. Answer: (a) ripples in space-time — They are space-time distortions.

17. Answer: (a) 2015 — LIGO detected on 14 Sept 2015.

18. Answer: (c) two black holes — Binary black hole merger.

19. Answer: (c) 2017 — GW170817 was in 2017.

20. Answer: (c) laser interferometry — It measures tiny arm length changes.

21. Answer: (c) 4 km — LIGO has 4 km arms.

22. Answer: (d) 3 km — Virgo has 3 km arms.

23. Answer: (d) Weiss, Barish and Thorne — LIGO leadership.

24. Answer: (a) LIGO — She worked on LIGO.

25. Answer: (c) spinning neutron stars — Steady spinning massive objects.

26. Answer: (d) orbiting compact objects — Black holes/neutron stars spiralling.

27. Answer: (a) violent events like supernovae — Short violent events.

28. Answer: (c) random background from many sources — Superposition of many weak signals.

29. Answer: (a) dimensionless — Strain = $\Delta L/L$.

30. Answer: (a) 10^{-21} — Extremely tiny fractional change.

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31. Answer: (a) 9.5 × 10^{12} km — Light travels about 9.5 trillion km in a year.

32. Answer: (a) 100 Hz — f = c/λ = $3\times10^8/3\times10^6$.

33. Answer: (c) 75 Hz — f = $3\times10^8/4\times10^6$.

34. Answer: (b) examine polarized light — It tests polarization.

35. Answer: (c) parallel — $\cos^2 0° = 1$.

36. Answer: (b) 0.25 — $\cos^2 60° = 0.25$.

37. Answer: (b) 0.15 — 100% − 85% = 15%.

38. Answer: (b) $I_0/2$ — Half the vibrations pass.

39. Answer: (b) 50 W/m² — $100\cos^2 45° = 50$.

40. Answer: (a) 8.66 units — $10\cos 30° \approx 8.66$.

41. Answer: (b) 56.3° — $\tan\theta = 1.5$.

42. Answer: (d) 53° — $\tan\theta = 1.33$.

43. Answer: (c) birefringent — It rotates polarization differently.

44. Answer: (c) dark — No scattering means no blue sky.

45. Answer: (c) almost unchanged — They interact very weakly.

46. Answer: (a) stretching and compressing — They distort objects.

47. Answer: (a) gravitational waves — But everyday waves are too tiny.

48. Answer: (b) laser interferometry — Interferometers measure strain.

49. Answer: (a) gravitational waves — Accelerating masses radiate GWs.

50. Answer: (c) transverse waves — It requires perpendicular vibrations.

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